

Přednáška 03

Databáze a vizualizační nástroje

Veřejně dostupná data měřených materiálů

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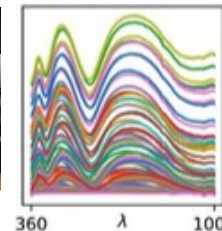
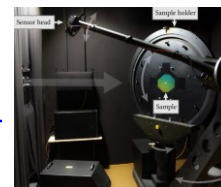
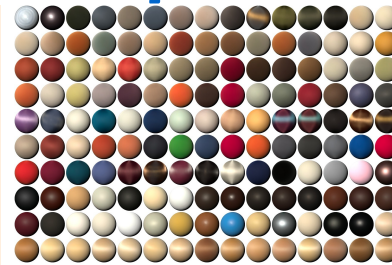
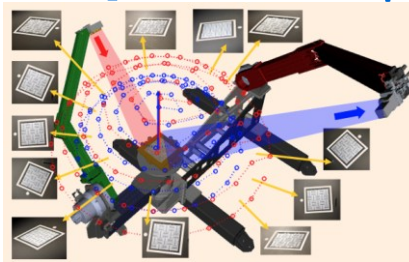
Czech Academy of Sciences



Appearance data publicly available

BRDF Databases

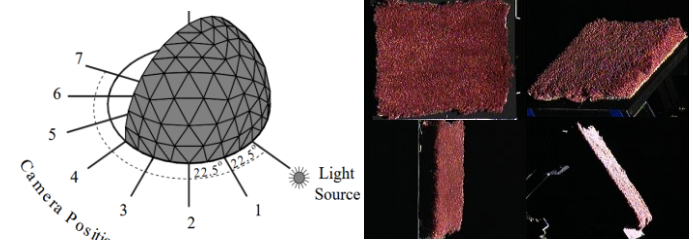
- MERL BRDF database [Matusik et al. SIG03]
 - 100 isotropic BRDFs
 - <https://merl.com/research/downloads/BRDF>
 - Fitted models: https://people.csail.mit.edu/addy/research/ngan05_brdf_supplemental_doc.pdf
- UTIA BRDF database [Vavra&Filip PG14] – 150 anisotropic BRDFs
 - <http://btf.utia.cas.cz>
- RGL BRDF database [Dupuy&Jacob 2018] – 36 spectral BRDFs (4 anisotropic), wavelengths 310-1100nm
 - <https://rgl.epfl.ch/publications/Dupuy2018Adaptive>



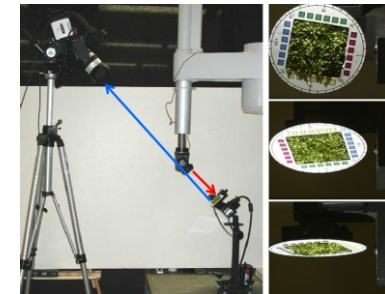
Appearance data publicly available

BTF Databases

- CURET-Columbia&Utrecht University
– 61 BTFs (limited sampling directions)
<http://www1.cs.columbia.edu/CAVE/software/curet>

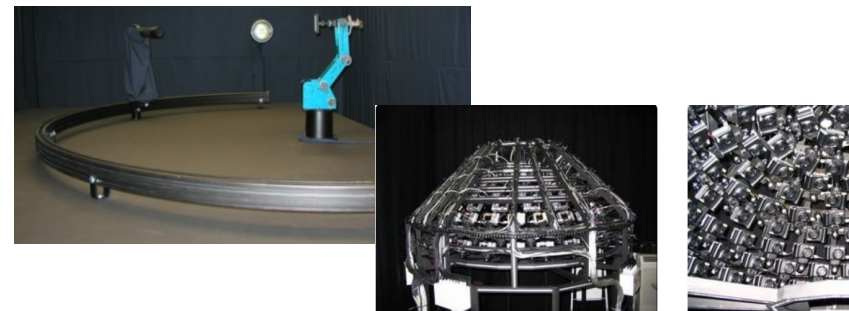


- Yale University BTF database
<http://vision.ucsd.edu/kriegman-grp/research/vst>



- University Bonn BTF database – 100 BTFs
<https://cg.cs.uni-bonn.de/data/btfdbb-btf-database-bonn-and-measurement-lab>

- UTIA BTF database – 22 BTFs
<http://btf.utia.cas.cz>



Appearance data publicly available

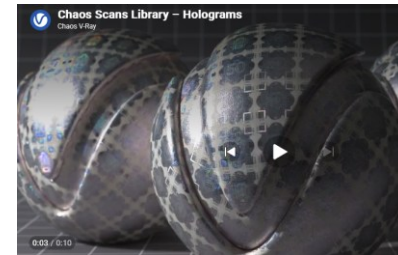
SVBRDF database

- OpenSVBRDF – 1000+ isotropic PBR maps
<https://opensvbrdf.github.io/>



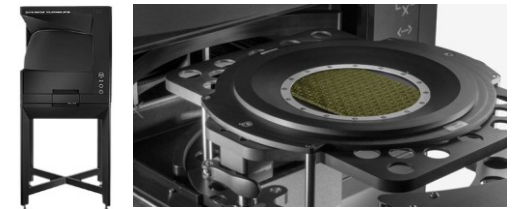
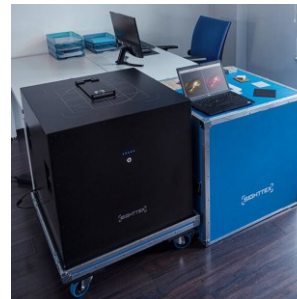
Commercial BTF scanning

- ChaosScans V-Ray – custom BTF format
<https://www.chaos.com/scans/features>



- TAC7 by xRite – custom BTF format
<https://www.xrite.com/categories/appearance/total-appearance-capture-ecosystem/tac7>

- SIGHTTEX Q by SightTex – open BTF formats (big, mif)
<https://www.sighttex.com/>



UTIA Anisotropic BRDF Database

<http://btf.utia.cas.cz>

6 carpet

96 fabric

- upholstery
- apparel
- cushion

10 other materials

plaster,
paper,
paint,

16 leather (genuine & imitations)

6 plastic

16 wood (genuine)

UTIA Anisotropic BRDF Database

<http://btf.utia.cas.cz>



elevation
step = 15°
azimuthal
step = 7.5°

UTIA BTF database – 22 materials



BTF database

- 6 BTFs as collection of images
- 16 BTFs in BIG data format

<http://btf.utia.cas.cz>

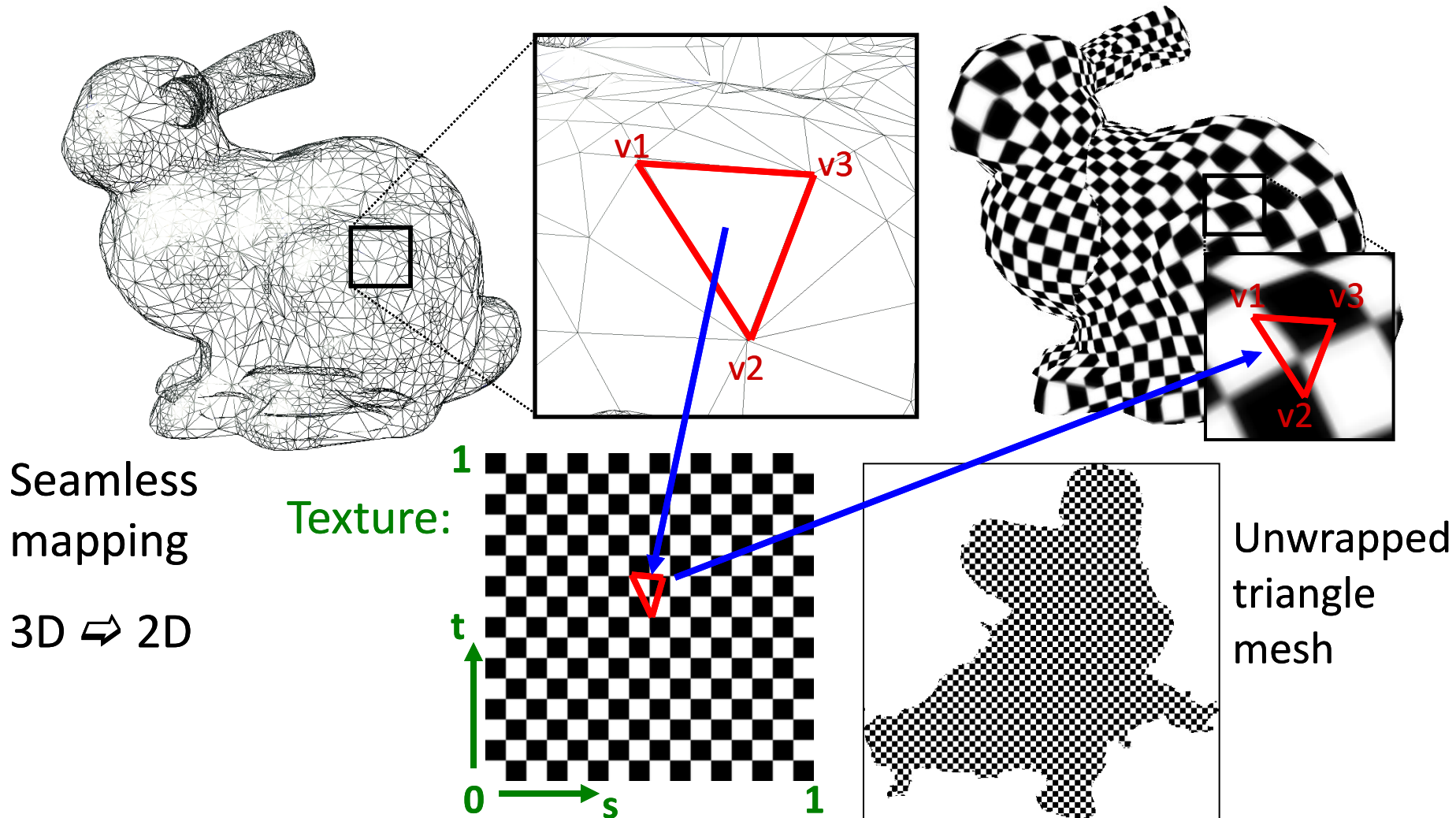
Outline - Appearance Visualization

- Our Goal
- Steps to achieve it:
 - Texture mapping
 - World vs. local coordinate system transformation
 - Local texture space
 - Local texture space interpolation
 - Measured directions interpolations
 - Point-light vs. environment illumination
 - Anti-aliasing



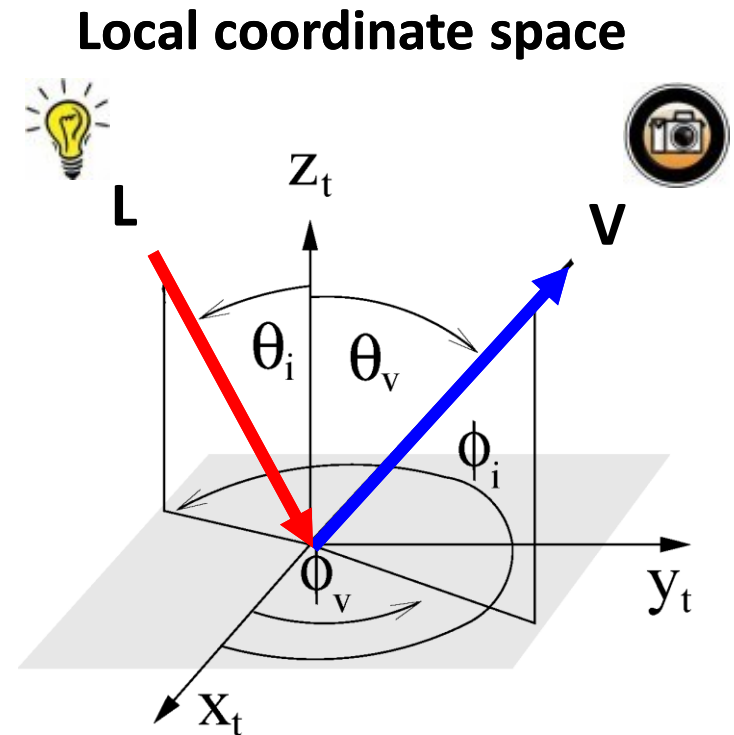
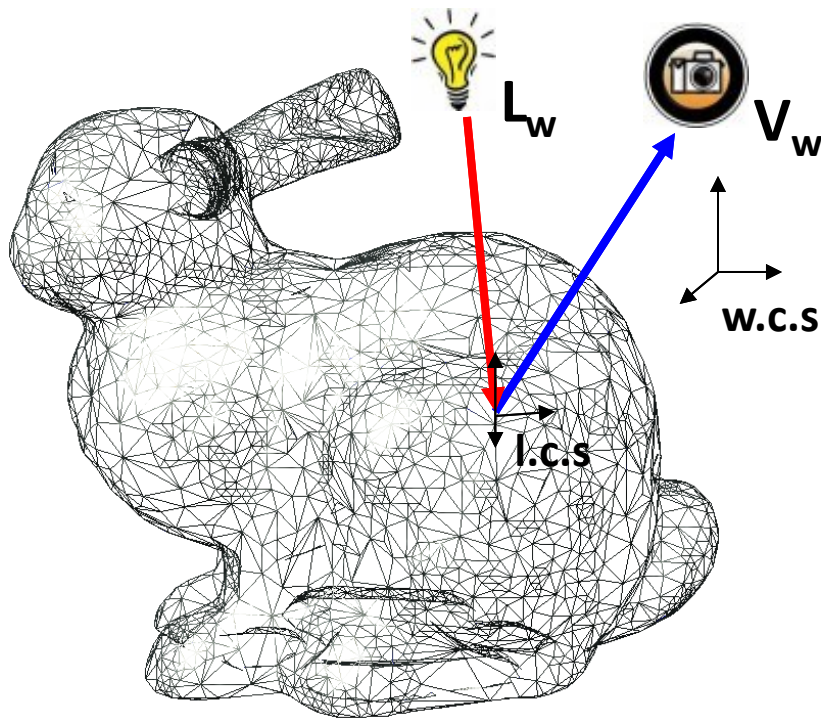
Texture Mapping

3D Geometry ($\mathbf{v}$) + Per-vertex 2D texture coordinates ($\mathbf{s}, \mathbf{t}$)



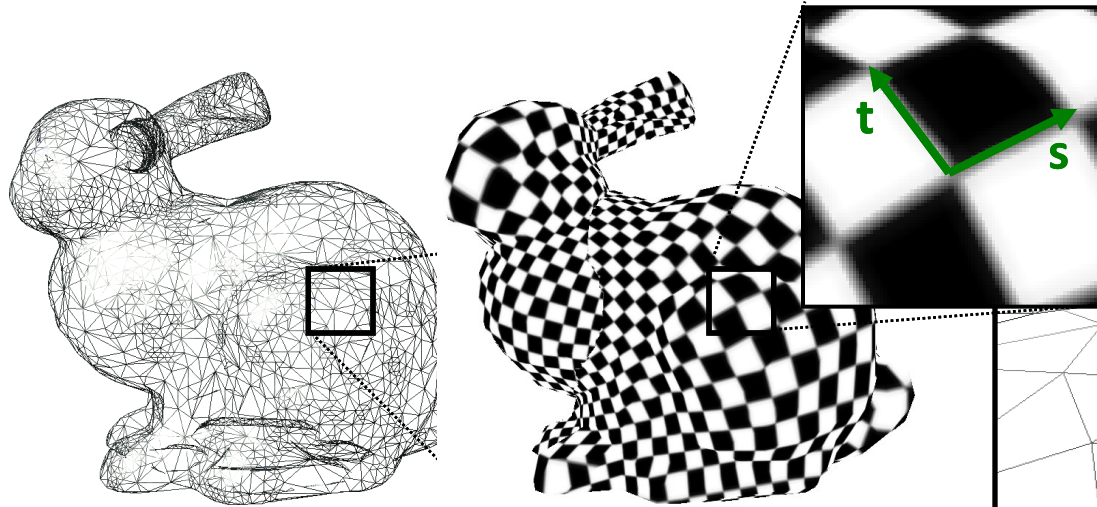
World vs. Local Coordinate System

$$RF(\lambda, r_1, r_2, \theta_i, \varphi_i, \theta_v, \varphi_v)$$



World light and camera positions - 3D world coordinates L_w V_w .
Coordinates in **local texture space** L , V ?

Local Texture Space



Geometry + texture mapping: $\mathbf{v}, \mathbf{s}, \mathbf{t}$

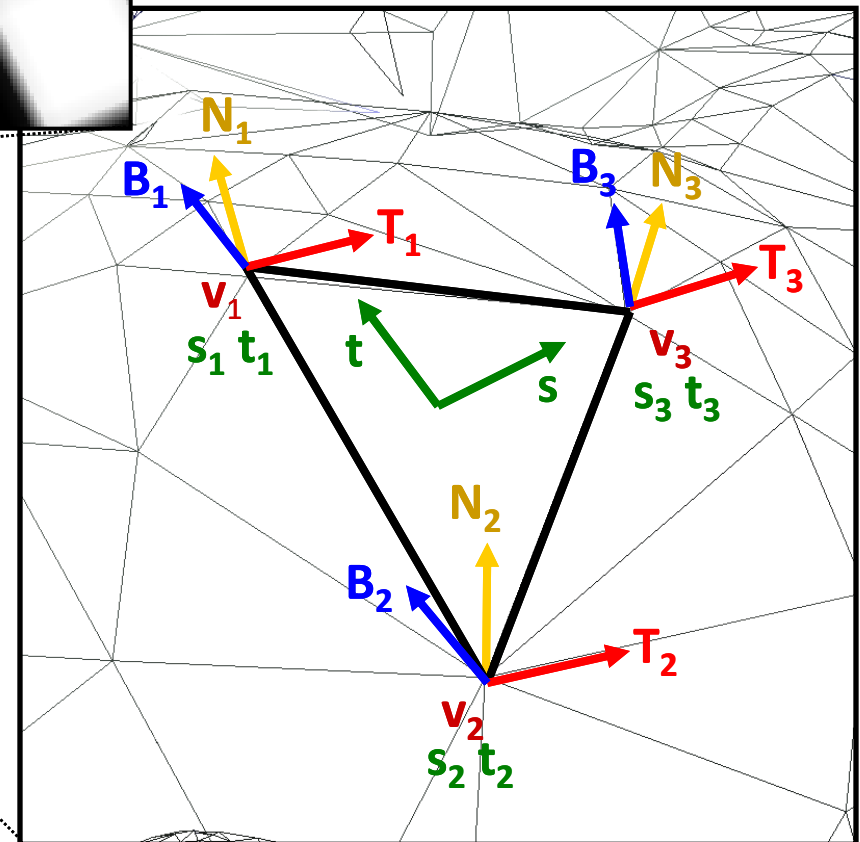
Local texture space:

$\mathbf{N}$... averaged per-face normals

$\mathbf{T}$... from values of $\mathbf{v}, \mathbf{s}, \mathbf{t}$

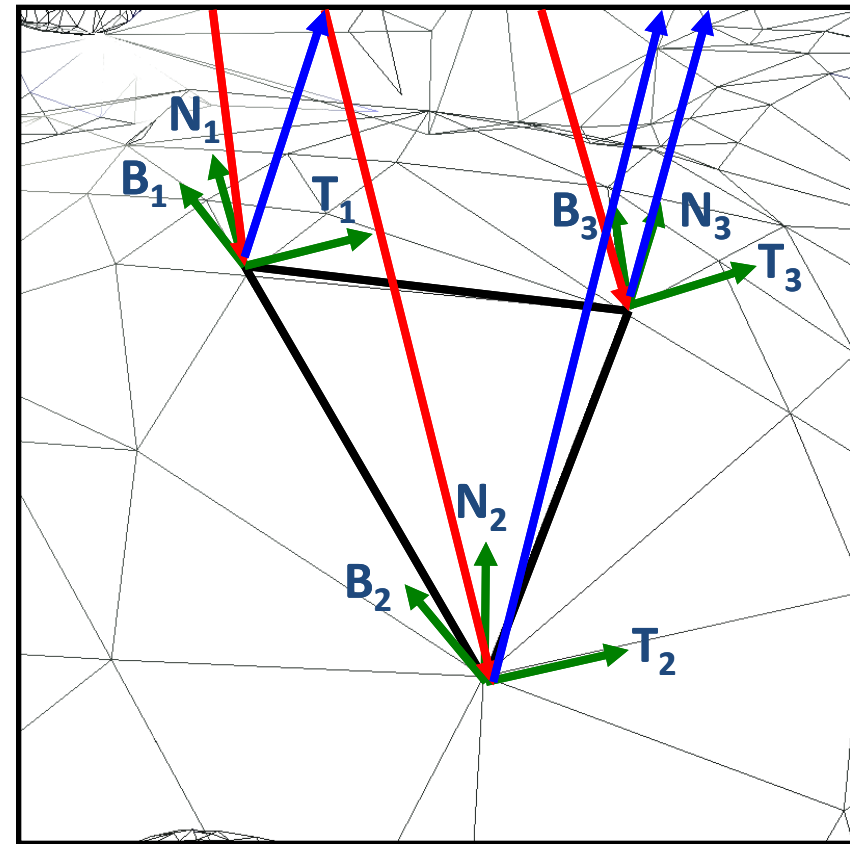
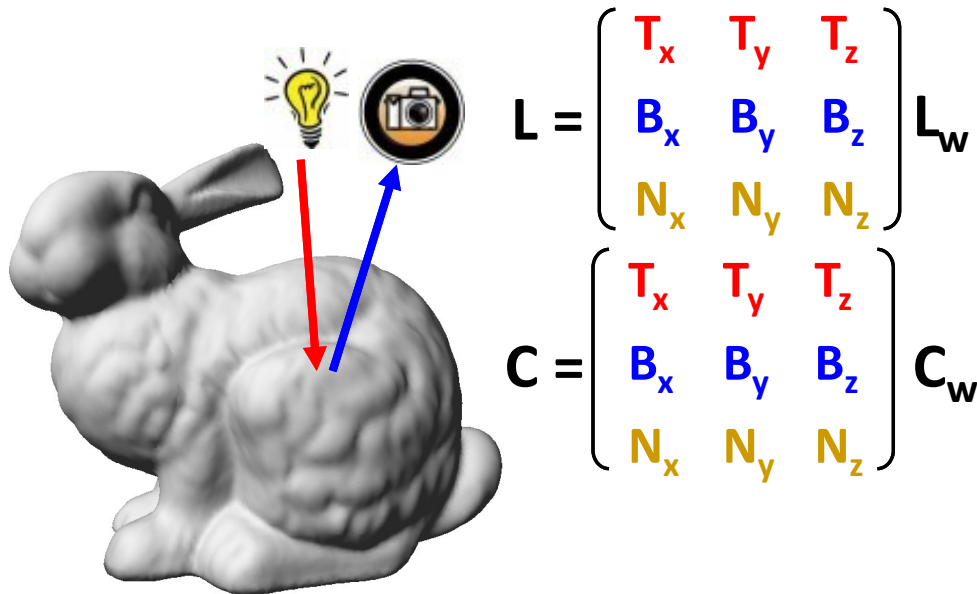
$\mathbf{B}$... crossproduct of $\mathbf{N}, \mathbf{T}$

$\mathbf{N}, \mathbf{T}, \mathbf{B}$ form system transforming
world $\Leftrightarrow$ local coordinates



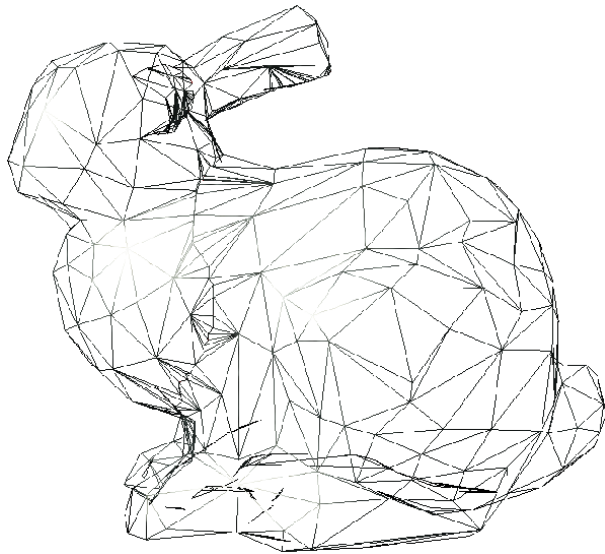
Local Texture Space

Obtaining illumination and view directions in local texture space:

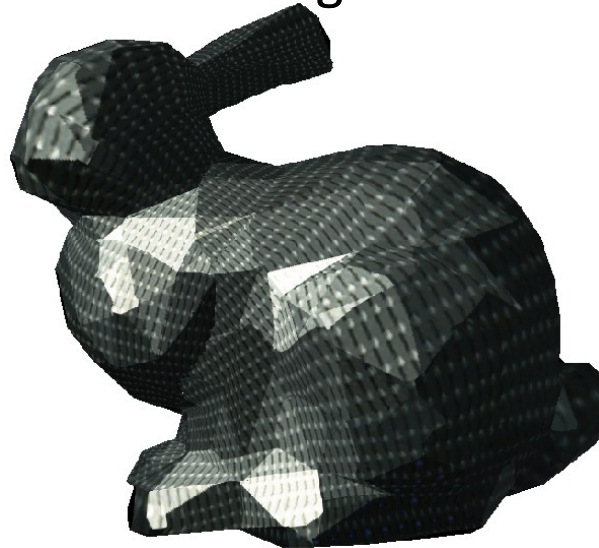


Local Texture Space (LTS) Interpolation

1000 triangles



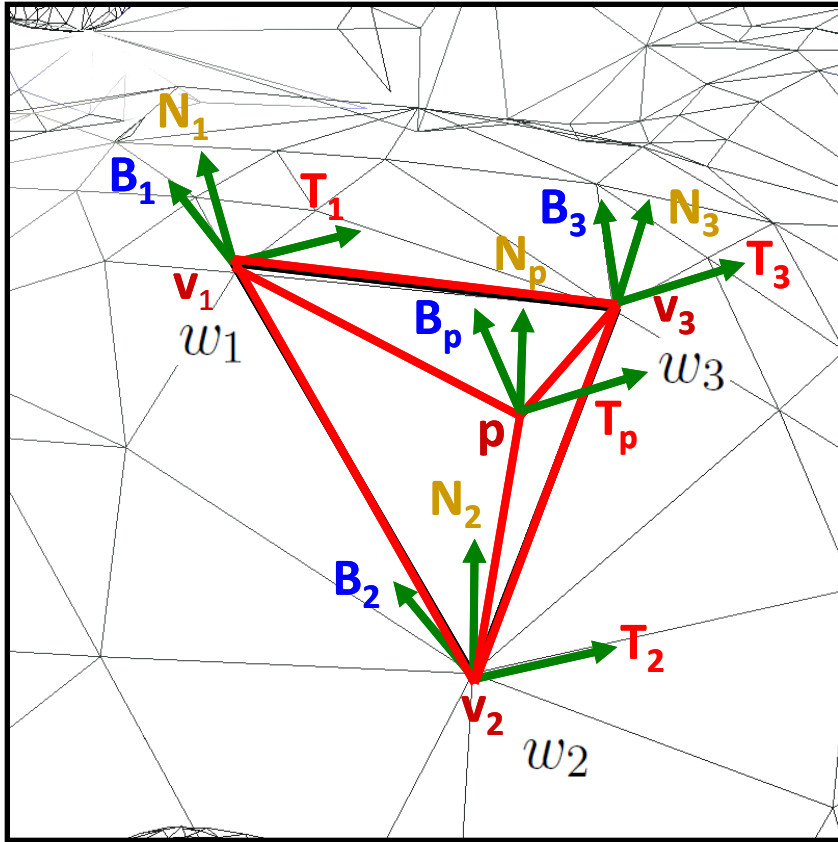
Per-triangle LTS



Per-pixel LTS



Local Texture Space Interpolation



Intra-triangle interpolation of **T**, **B**, **N** is supported in shaders of graphics HW.

Barycentric interpolation obtains weights as a ratios of corresponding triangles areas.

$$w_1 = \frac{\Delta(\mathbf{p}, \mathbf{v}_2, \mathbf{v}_3)}{\Delta(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)}$$

$$w_3 = \frac{\Delta(\mathbf{p}, \mathbf{v}_1, \mathbf{v}_2)}{\Delta(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)}$$

$$w_2 = \frac{\Delta(\mathbf{p}, \mathbf{v}_1, \mathbf{v}_3)}{\Delta(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)}$$

$$w_1 + w_2 + w_3 = 1$$

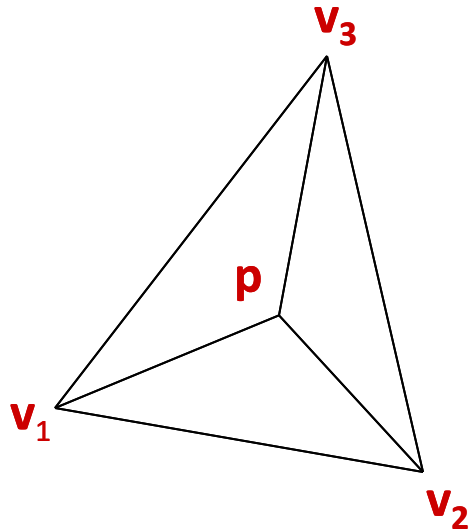
$$\mathbf{T}_p = w_1 \mathbf{T}_1 + w_2 \mathbf{T}_2 + w_3 \mathbf{T}_3$$

$$\mathbf{B}_p = w_1 \mathbf{B}_1 + w_2 \mathbf{B}_2 + w_3 \mathbf{B}_3$$

$$\mathbf{N}_p = w_1 \mathbf{N}_1 + w_2 \mathbf{N}_2 + w_3 \mathbf{N}_3$$

Barycentric Interpolation

Mobius 1827 [Coxeter 1969, Fauvel et al. 1999]



$$w_1 = \frac{\|(\mathbf{v}_2 - \mathbf{p}) \times (\mathbf{v}_3 - \mathbf{v}_2)\|}{\|(\mathbf{v}_2 - \mathbf{v}_1) \times (\mathbf{v}_3 - \mathbf{v}_2)\|}$$

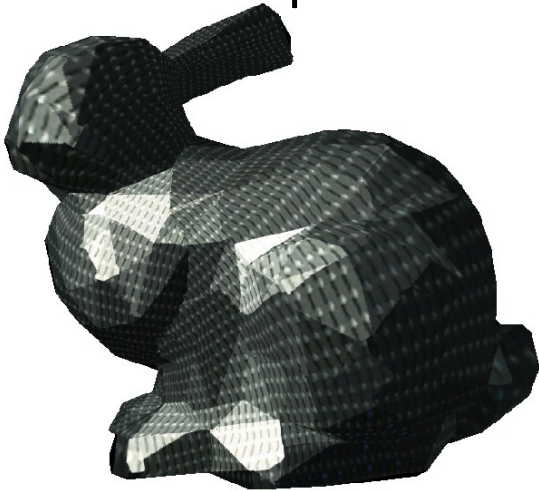
$$w_2 = \frac{\|(\mathbf{v}_3 - \mathbf{p}) \times (\mathbf{v}_1 - \mathbf{v}_3)\|}{\|(\mathbf{v}_2 - \mathbf{v}_1) \times (\mathbf{v}_3 - \mathbf{v}_2)\|}$$

$$w_3 = \frac{\|(\mathbf{v}_1 - \mathbf{p}) \times (\mathbf{v}_2 - \mathbf{v}_1)\|}{\|(\mathbf{v}_2 - \mathbf{v}_1) \times (\mathbf{v}_3 - \mathbf{v}_2)\|}$$

- If w_1, w_2 and w_3 are all greater than zero, $\mathbf{p}$ is strictly inside the triangle.
- If $w_i = 0$ and the other two coordinates are positive, $\mathbf{p}$ lies on the edge opposite $\mathbf{v}_i$.
- If $w_i = 0$ and $w_j = 0$, $\mathbf{p}$ lies on $\mathbf{v}_k$.
- If $w_i < 0$, $\mathbf{p}$ lies outside the edge opposite $\mathbf{v}_i$.

Measured Directions Interpolation

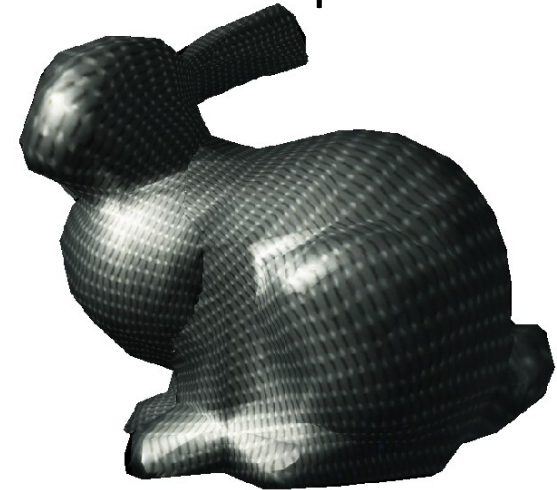
NO interpolation



LTS interpolation

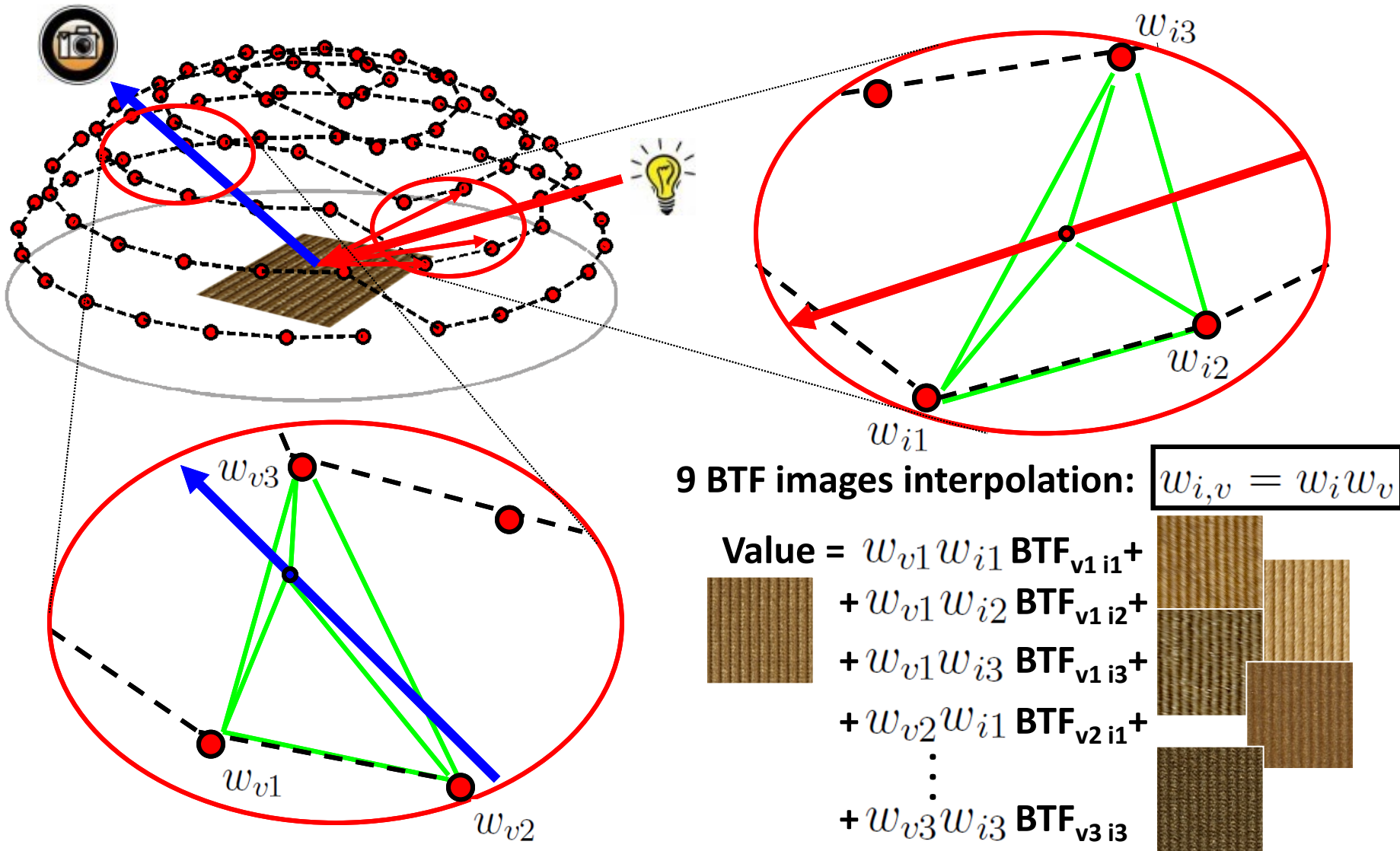


LTS+BTF interpolation



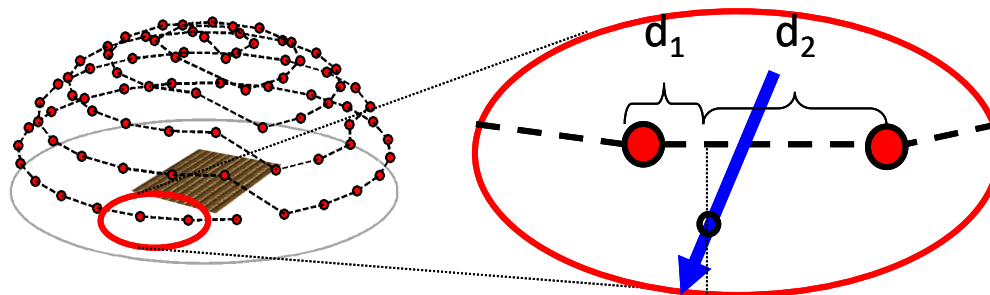
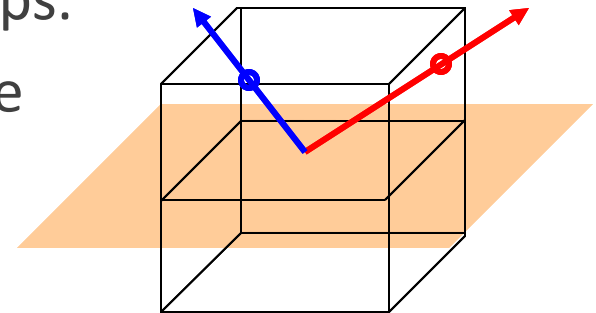
Not needed by analytical models $\Rightarrow$ reflectance for arbitrary directions

Measured Directions Interpolation



Measured Directions Interpolation

- Interpolation $\Rightarrow$ for each pixel (fragment) of rendered object,
- 3 closest directions indices & interpolation weights can be precomputed and stored, e.g. in cube-maps.
- Easy implementation in graphics hardware
- For elevation angles higher than those measured, a modification is required:



Appearance Rendering

- Total **material radiance** $\Rightarrow$ emitted + reflected:

$$L_r(x, y, \omega_v) = \cancel{L_{em}(x, y, \omega_v)} + L_{rf}(x, y, \omega_v)$$

- General **rendering equation**

$$L_r(x, y, \omega_v) = \int_{\Omega_i} \underbrace{BRDF(\omega_i, \omega_v)}_{\text{Surface reflectance}} \underbrace{L_i(x, y, \omega_i)}_{\text{Illumination contribution}} \underbrace{\cos(\theta_i)}_{\text{Illuminated area foreshortening}} d\omega_i$$

Surface
reflectance

Illumination
contribution

Illuminated area
foreshortening

Illum./view
direction

Appearance Rendering

- Reflected **radiance integration over hemisphere**

- Foreshortening term comprised in BTF data

$$L_r(x, y, \omega_v) = \int_{\Omega_i} BTF(x, y, \omega_i, \omega_v) L_i(x, y, \omega_i) d\omega_i$$

- Impossible to solve analytically $\Leftrightarrow$

- Rendering for a **single directional-light**

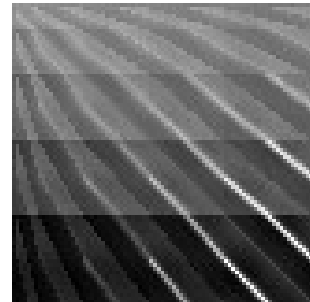
- $\Leftrightarrow$ unrealistic

$$L_r(x, y, \omega_v) = BTF(x, y, \omega_i, \omega_v) L_i(x, y, \omega_i)$$

- Approximation by a **finite number of directional lights**

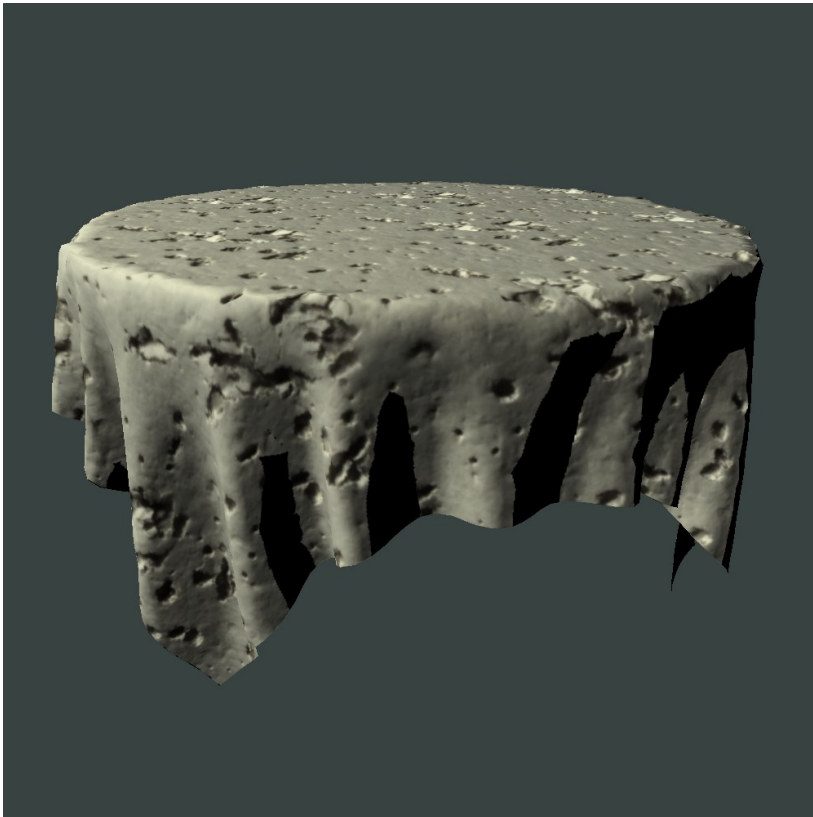
$$L_r(x, y, \omega_v) = \sum_i^{n_i} BTF(x, y, \omega_i, \omega_v) L_i(x, y, \omega_i)$$

Mean pixel
luminance –
leather BTF



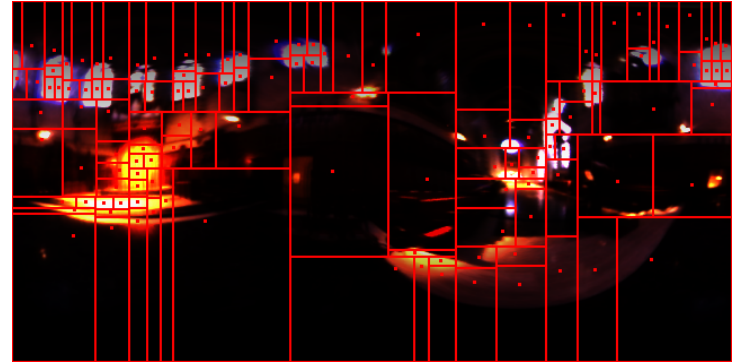
Point-Light vs. Environment Illumination

- Point-light
 - Idealized, non-realistic
 - Fast computation
- Environment Illumination
 - Realistic
 - More demanding comp.

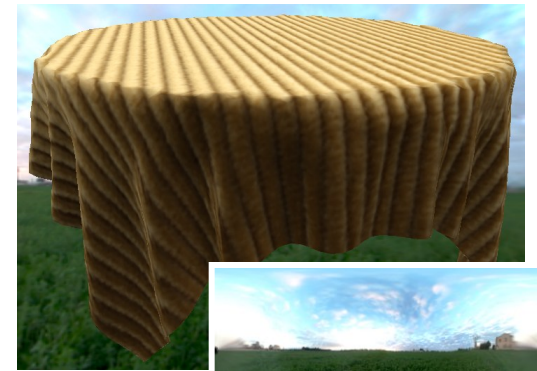
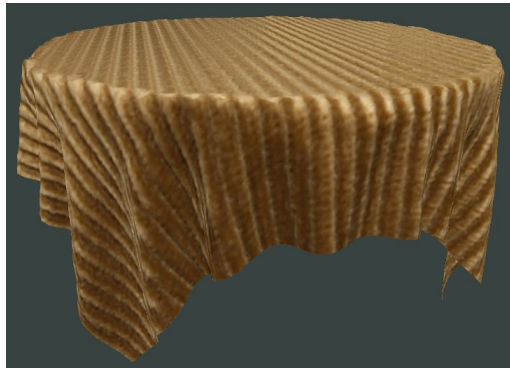


Environment Illumination

- Environment approximated by a set of point-lights

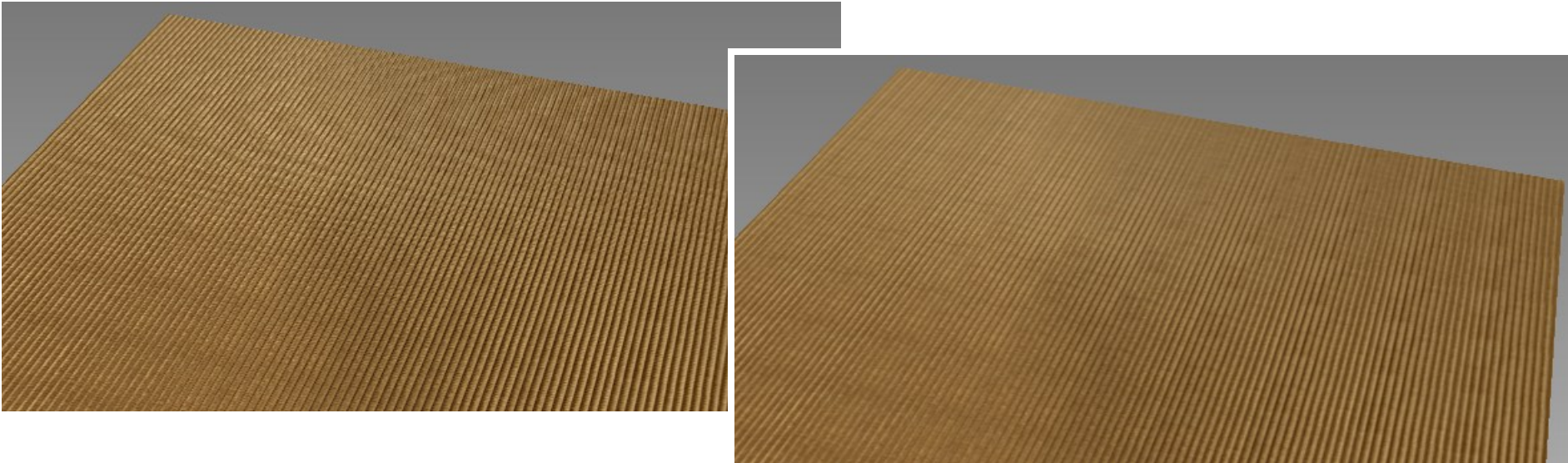


- Smoother images - convolution of pixel with lights
- Slower rendering - optimization techniques are available



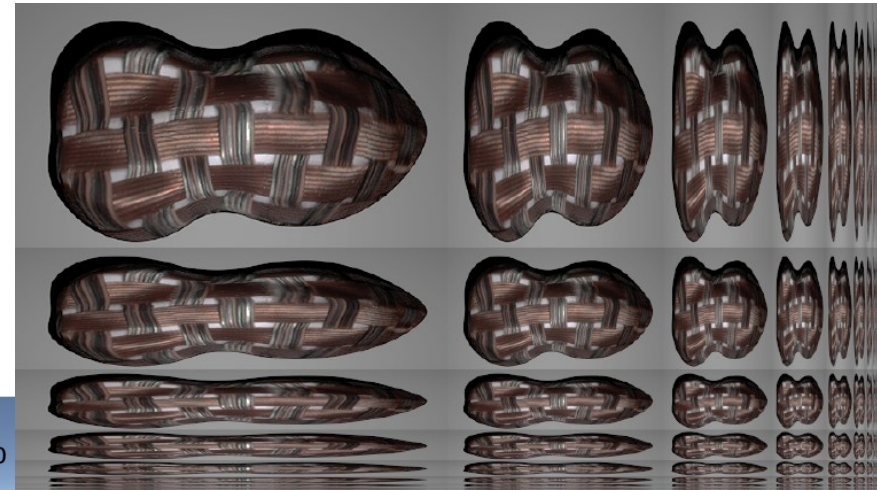
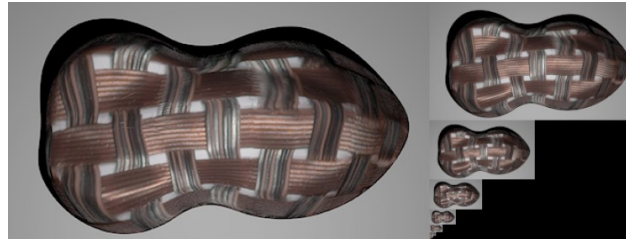
Anti-Aliasing

- Artifacts caused by image rasterization
- Mip-mapping
 - image down-sampling to lower resolutions (Gaussian pyramid)
 - resolution selected (pyramid levels) from observer distance



Anti-Aliasing

- Mip-mapping
 - 50% overhead data
- Anisotropic filtering
 - 100% overhead data

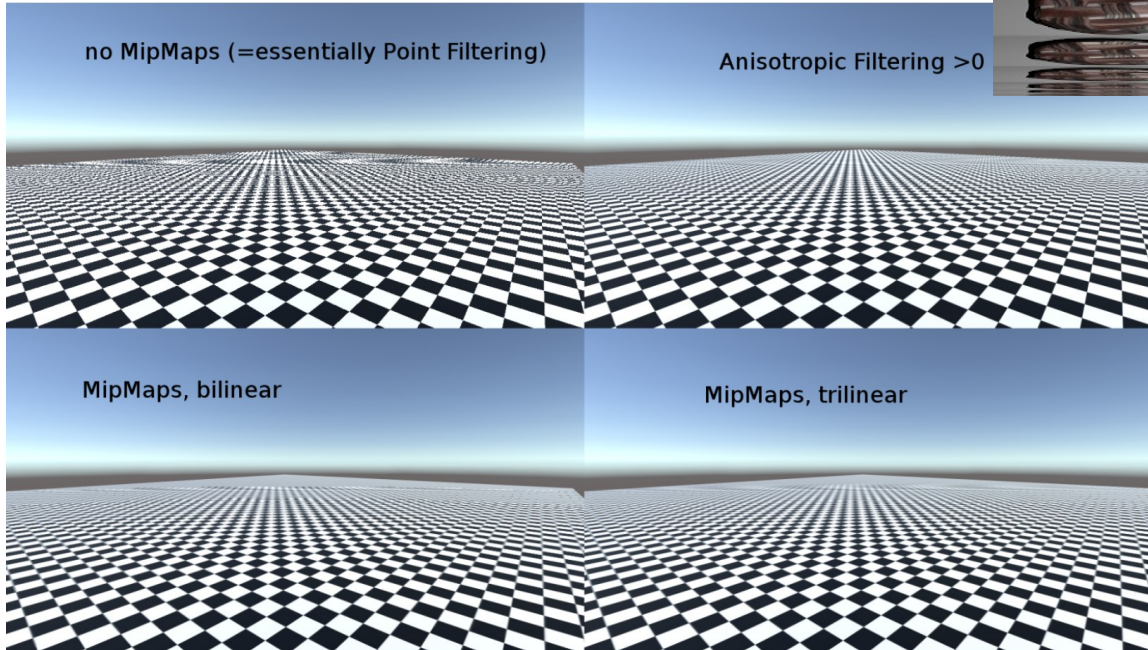


no MipMaps (=essentially Point Filtering)

Anisotropic Filtering >0

MipMaps, bilinear

MipMaps, trilinear



Implementation of Appearance Visualization

- Some processing steps can (has) to be precomputed:
 - Texture mapping
 - Per-vertex Local texture space (LTS) on 3D object
 - Measured directions Barycentric interpolation
- Other have to be computed per-pixel on the fly:
 - Conversion illumination and view directions to LTS
 - LTS interpolation across triangles (HW supported)
 - reflectance model evaluation (BRDF, SVBRDF, BTF, ...)
 - Environment illumination, anti-aliasing
- Benefit from parallelized graphics hardware
 - Shading languages: OpenGL+GL Shading Language, CgFX,
 - Higher level programming: CUDA, OpenCL

Visualization tools

- BRDF Explorer (Disney)

https://www.songho.ca/opengl/gl_disney.html

- Blender

<https://www.blender.org/>

- SIGHTTEX Studio

<https://www.sighttex.com/webform/download-sighttex-studio-free>

- MITSUBA 2.0 (C++)

<https://github.com/mitsuba-renderer/mitsuba2>

- MITSUBA 3.0 (python)

<https://github.com/mitsuba-renderer/mitsuba3>

